# IIT Gandhinagar

## Week 2 Schedule | Data Analysis with Pandas and SQL

Hello Everyone,
Please find below the schedule for Week 2. We will focus on solving business problems using Pandas and SQL, analysing data, and presenting meaningful insights. All timings are in IST.

### Day 1: Cohort Analysis and Joins

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| **10:00 – 11:00 AM** | Inaugural Session – IIT Director addressing the cohort |
| **11:00 AM – 12:00 PM** | Student Presentations and Discussion |
| **12:00 – 1:00 PM** | Cohort Analysis using Pandas GroupBy |
| **2:00 – 4:00 PM** | IIT Faculty Session – Prof. Mithun: Sampling Distributions and Central Limit Theorem |
| **4:30 – 6:30 PM** | Pandas GroupBy, Pandas Joins and SQL Joins |

### Day 2: Distributions and Renewal Analysis

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| **10:00 AM – 1:00 PM** | Distribution, Process and Funnel Analysis using Pandas |
| **2:00 – 4:00 PM** | IIT Faculty Session – Prof. Mithun: Confidence Intervals and t-Tests |
| **4:30 – 6:30 PM** | SQL GROUP BY and Pandas GroupBy Practice |

### Day 3: Fair Comparisons and Benchmarking

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| **10:00 AM – 12:00 PM** | Student Presentations and Discussion |
| **12:00 – 1:00 PM** | Benchmarking and Standardisation using Pandas |
| **2:00 – 4:00 PM** | IIT Faculty Session – Prof. Mithun: Errors, Statistical Power, Sample Size and Multiple Comparisons |
| **4:30 – 6:30 PM** | SQL CASE Statements, Date Functions and Pandas Date Operations |

### Day 4: Customer and Product Analysis

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| **10:00 AM – 12:00 PM** | RFM Analysis and Market Basket Analysis using Pandas |
| **12:00 – 1:00 PM** | Pandas Practice |
| **2:00 – 5:00 PM** | SQL and Pandas Practice |

### Day 5: Cost Analysis and Business Problem-Solving

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| **10:00 – 11:00 AM** | Student Presentations and Discussion |
| **11:00 AM – 1:00 PM** | Cost-to-Serve Analysis using Pandas and SQL |
| **2:00 – 5:00 PM** | Detailed Pandas and SQL Exercises |

### Key Topics Covered This Week

Cohort and funnel analysis • Pandas GroupBy, aggregation and joins • SQL joins, GROUP BY, CASE statements and date functions • Statistical distributions and hypothesis testing • Benchmarking and fair comparisons • RFM and basket analysis • Cost-to-serve analysis and business recommendations

**Important:** Please be on time and actively participate in discussions, presentations and hands-on practice sessions.
**Let's have a productive week of learning and problem-solving!**